Unit 6:
WAN, Network Management, and Enterprise Networks

6.2 Network Automation, Cloud, IoT, Enterprise Design & Zero Trust

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Learning Objectives



By the end of this lesson, students will be able to:

✅ Explain why we automate networks and what Python scripts do
✅ Describe SDN as separating the "brain" from the "muscles"
✅ Compare IaaS, PaaS, SaaS and understand VPC
✅ Name three IoT protocols and the edge/fog/cloud layers
✅ Design a simple hierarchical network with redundancy
✅ State the main idea of Zero Trust: "Never trust, always verify"

What is Network Automation?

Definition: Using software and scripts to configure, manage, and operate network devices automatically – without typing commands manually on each device.

Why now?

Networks are getting bigger and more complex. Manual work doesn't scale.

Analogy: Washing dishes by hand vs. using a dishwasher – both work, but automation saves time and reduces mistakes.

The Problem with Manual Network Management

Manual way:

- Log into each router/switch one by one (SSH or console)
- Type the same commands again and again
- Risk typos (e.g., wrong VLAN number)
- No record of who changed what

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Real-world example: A company with 200 switches. A security update requires changing a password on all of them. Manual = 200 logins (~4 hours). One typo = outage.

How Automation Solves the Problem

Automated way:

- Write one script or use an automation tool
- Run it once – it touches all 200 devices
- Every device gets exactly the same configuration
- Changes are logged and can be undone easily

Time comparison: 4 hours manual → 30 seconds automated.

Analogy: Manual = handwriting 1000 letters. Automation = mail merge.

Python for Network Automation

Why Python?

- Easy to learn (reads like English)
- Has many libraries for networking
- Used by most network engineers

Python is not just for programmers – many network engineers learn basic Python to save time.

Python Libraries for Network Automation

A library is a collection of pre-written code that does specific tasks.

Library	What it does
netmiko	Logs into devices via SSH and runs commands (Cisco, Juniper, etc.)
napalm	Works with many vendors using the same commands
paramiko	Lower-level SSH (when netmiko isn't enough)
requests	Talks to REST APIs (cloud, SDN controllers)

Analogy: Libraries are like toolkits – you don't build a hammer, you just use one.

What Can You Automate? (Real Tasks)

Task	Manual time	Automated time
Backup configs of 100 switches	2+ hours	1 minute
Push new VLAN to 50 switches	1 hour	30 seconds
Check all devices for a specific setting	3 hours	1 minute
Reboot 20 routers at 2 AM	(stay awake)	Scheduled script

Result: Network engineers do less boring work and more interesting problem-solving.

Automation Benefits – Top 5

1. **Speed** – Changes happen in seconds, not hours
2. **Consistency** – No "I forgot to save the config"
3. **Compliance** – Every device matches the company standard
4. **Auditing** – Git tracks every change (who, when, what)
5. **Self-healing** – Scripts can detect failures and fix them automatically

Analogy: Automation turns a network from a hand-cranked machine into a modern robot.

SDN – Software-Defined Networking

A new way to build networks where the **control plane** (brain) is separated from the **data plane** (muscles).

Traditional network: Every switch/router has its own brain. They talk to each other but each decides independently.

SDN: One central brain (controller) tells all switches exactly what to do.

Analogy: Traditional = each taxi driver chooses their own route. SDN = a dispatch center tells all taxis where to go.

Control Plane vs. Data Plane

Plane	What it does	Where it lives in traditional networking	Where it lives in SDN
Control plane	Makes decisions (routing, policies)	On each switch/router	Centralized controller
Data plane	Forwards packets based on decisions	On each switch/router	Still on switches (but follows controller's orders)

Simple Analogy: Control plane = brain (thinks). Data plane = hands (does). SDN takes the brain out of each device and puts it in one place.

OpenFlow – The Language SDN Uses

OpenFlow is the protocol that allows the SDN controller to talk to switches.

The controller sends rules to the switch:

- **Match:** Look for packets with specific characteristics (source IP, destination IP, port, etc.)
- **Action:** Do something with those packets (forward, drop, modify, send to controller)
- **Counters:** Keep track of how many packets/bytes matched

Example rule:

Match: source IP = 10.0.0.5, destination IP = 8.8.8.8, protocol = TCP

Action: forward out port 4

Counter: +1 packet, +64 bytes

SDN Controllers – The Brains

The **SDN controller** is the software that runs the control plane.

Popular SDN controllers:

- OpenDaylight (open source, very popular)
- ONOS (open source, focused on service providers)
- Ryu (simple, good for learning)

What a controller does:

- Learns the network topology (which switches are connected)
- Installs flow rules on switches via OpenFlow
- Responds to changes (link down, new device)

Analogy: The controller is like an air traffic controller – pilots (switches) follow instructions.

Cloud Computing – What Is It?

Cloud computing = using computing resources (servers, storage, databases, networking) over the internet, paying only for what you use.

Instead of buying your own servers:

- You rent them from a cloud provider (AWS, Microsoft Azure, Google Cloud)
- You can increase or decrease capacity in minutes

Analogy: Buying a car vs. using Uber.
Owning a car = traditional data center. Uber = cloud – pay per trip, no maintenance.

Cloud Computing – Three Service Models

Model	What you manage	Example	Analogy
IaaS	Apps, data, OS, network	AWS EC2 (virtual servers)	Rent an empty apartment
PaaS	Apps, data only	Google App Engine (deploy code)	Rent a furnished apartment
SaaS	Nothing – just use it	Gmail, Office 365	Stay in a hotel

Simple rule: More control = more responsibility (IaaS). Less control = less work (SaaS).

IaaS – Infrastructure as a Service

What you get: Virtual machines, storage, networking, firewalls, load balancers – all in the cloud.

You are responsible for: The operating system, applications, data, security patches (on the VM).

Example: A company needs a web server. They launch an EC2 instance on AWS, install Apache, and configure it.

Best for: Companies that want full control but don’t want to buy physical servers.

PaaS – Platform as a Service

What you get: A platform to develop and run applications. The provider manages the OS, runtime, and infrastructure.

You are responsible for: Upload your code. The platform runs it.

Example: A developer writes a Python web app and deploys it to Google App Engine. She never touches the server.

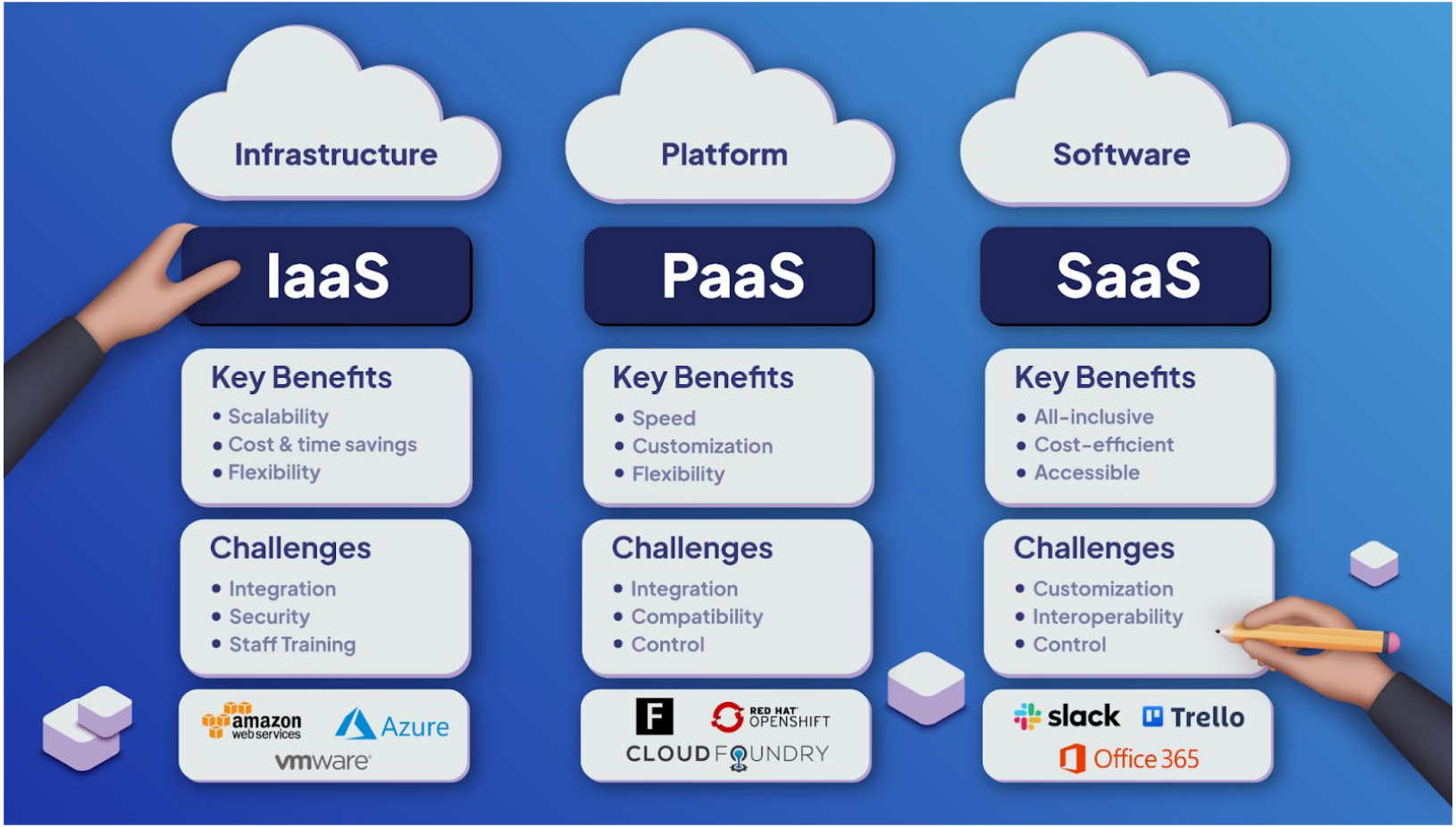
Best for: Developers who just want to write code without managing servers.

SaaS – Software as a Service

What you get: A complete software application that runs in your web browser.
You don't: Install, update, or manage anything.

Example: Gmail, Google Docs, Salesforce, Zoom, Microsoft 365.

Best for: End users who just want to use software.
For companies: No IT staff needed to maintain the software – just pay a subscription.


Infrastructure
Platform
Software
IaaS
PaaS
SaaS
Key Benefits
• Scalability
• Cost & time savings
• Flexibility
Key Benefits
• Speed
• Customization
• Flexibility
Key Benefits
• All-inclusive
• Cost-efficient
• Accessible
Challenges
• Integration
• Security
• Staff Training
Challenges
• Integration
• Compatibility
• Control
Challenges
• Customization
• Interoperability
• Control
amazon web services
Azure
vmware
RED HAT OPENSHIFT
CLOUD FOUNDRY
slack
Trello
Office 365

VPC – Virtual Private Cloud

A logically isolated section of a public cloud that is dedicated to one customer.

What you can do in a VPC:

- Choose your own private IP range (e.g., 10.0.0.0/16)
- Create subnets (e.g., 10.0.1.0/24 for web servers, 10.0.2.0/24 for databases)
- Define security groups (firewall rules)
- Connect to your on-premises network

Analogy: A gated community inside a large city – you have your own streets, gates, and security.

Connecting Your Office to a VPC

Method	How it works	Speed	Security	Cost
VPN over internet	Encrypted tunnel through the public internet	Depends on your internet (variable)	Good (AES encryption)	Low
Direct Connect / ExpressRoute	Private dedicated circuit from your office to cloud provider	1 Gbps – 10 Gbps (dedicated)	Excellent (private, not over internet)	High

When to choose:

- Small office or temporary connection → VPN
- Large enterprise or sensitive data → Direct Connect

Hybrid Cloud

Definition: A mix of on-premises (your own data center) and public cloud, with them connected securely.

Why do this?

- Data sovereignty: Some data must stay on-prem (laws, compliance)
- Cloud bursting: Normal traffic on-prem, extra traffic goes to cloud during peaks
- Disaster recovery: If your data center fails, cloud takes over

Example: A retail website – normal days run on company servers; Black Friday traffic automatically goes to AWS.

IoT – Internet of Things

IoT = Connecting everyday physical objects to the internet so they can send and receive data.

Examples you may know:

- Smart thermostat (Nest)
- Fitness tracker (Fitbit)
- Smart light bulbs
- Voice assistants (Amazon Echo)
- Industrial sensors on machines

Analogy: Giving a voice to physical objects – your fridge can tell you when you're out of milk.

What Makes IoT Different?

Aspect	Regular computer	IoT device
Power	Plugged in or large battery	Tiny battery (lasts months/years)
CPU	Fast (GHz)	Slow (MHz)
Memory	GBs of RAM	KBs or MBs of RAM
Network	Ethernet, Wi-Fi	Often low-power radio (Zigbee, LoRa)
Scale	Dozens or hundreds	Thousands or millions

Result: IoT needs special, lightweight protocols.

IoT Protocols – MQTT, CoAP, Zigbee

Protocol	Transport	Style	Best for
MQTT	TCP	Publish/subscribe – sensors publish data, apps subscribe	Home automation, sensor networks
CoAP	UDP	Request/response (like HTTP but smaller)	Very constrained devices (8-bit microcontrollers)
Zigbee	Radio (802.15.4)	Mesh networking – devices relay for each other	Smart lighting, industrial control

Simple rule:

- Internet-connected IoT → MQTT or CoAP
- Local mesh of devices → Zigbee

IoT Architecture – Edge, Fog, Cloud

Layer	Where it runs	What it does	Speed	Example
Edge	On the IoT device itself	Sense data, act immediately (no network needed)	Milliseconds	A smoke detector sounds alarm locally
Fog	Local gateway, router, or edge server	Collects data from many devices, filters, responds locally	Sub-second	A factory gateway shuts down a machine if temperature is too high
Cloud	Remote data center	Long-term storage, complex analytics, AI training	Seconds to minutes	All sensor data stored for monthly reports, predictive maintenance

Analogy: Edge = your reflexes (instant). Fog = your spinal cord (very fast). Cloud = your brain (slower but smarter).

Enterprise Network Design – Three Layers

Large networks are built in three layers (hierarchical design):

Layer	Role	Devices
Core	High-speed backbone – just moves traffic fast	Big routers/switches
Distribution	Applies policies, routing between VLANs	Layer 3 switches
Access	Connects end devices (PCs, printers, phones)	Layer 2 switches

Analogy: Core = interstate highway, Distribution = city ring road, Access = neighborhood streets.

Why Three Layers?

- Modular – You can upgrade the access layer without touching the core.
- Scalable – Add more access switches easily.
- Fault isolation – A problem in the access layer doesn’t crash the whole network.

Result: Easier to manage, troubleshoot, and grow.

Redundancy – No Single Point of Failure

Redundancy means having backups so that one failure doesn't stop everything.

Types:

- Device redundancy – Two routers with HSRP/VRRP (one takes over if the other fails)
- Link redundancy – Multiple cables with EtherChannel/LACP
- Power redundancy – Two power supplies, UPS (battery backup)

Uptime measurement – "Nines":

- 99% → 3.65 days downtime/year
- 99.9% → 8.76 hours
- 99.999% → 5 minutes (that's excellent)

Load Balancing & High Availability

- Load balancing: Spreading traffic across multiple servers so no single server gets overwhelmed.
  - Layer 4 (IP/port) – Simple, fast
  - Layer 7 (HTTP content) – Smarter, can look at URLs

- High Availability (HA):
  - Active-Passive: One server works, the other waits (standby)
  - Active-Active: Both work, sharing the load

- Example: A popular website uses a load balancer in front of 10 web servers – if one server crashes, users never notice.

Zero Trust Security – The Old Way vs. New Way

Old way (castle-and-moat):

- Trust everything inside your network
- Assume outsiders are bad, insiders are good

Problem: Once an attacker gets inside, they can move freely.

Zero Trust:

- Never trust anyone, even inside
- Always verify every request
- "Never trust, always verify"

Analogy: Airport security at every door, not just at the entrance.

Zero Trust Security



Five Principles of Zero Trust

1. **Never trust, always verify** – No automatic trust based on location
2. **Least privilege** – Give only the minimum access needed
3. **Assume breach** – Design as if an attacker is already inside
4. **Micro-segmentation** – Break the network into tiny isolated pieces
5. **Continuous verification** – Keep checking, not just once

Result: If one part is hacked, the attacker can’t easily move to other parts.

Identity-Based Access Control

Traditional access: Based on IP address or network location (e.g., "inside the office").

Zero Trust access: Based on who you are, your device health, location, time, behavior.

Example:

- Normal: Employee at office 2 PM → access granted
- Suspicious: Same employee at coffee shop 2 AM → ask for extra MFA or block

Analogy: A nightclub that checks ID at the door AND every time you enter a new room.

Micro-Segmentation & Continuous Monitoring

Micro-segmentation: Creating very small network segments – each with its own security policy.

Example – three-tier web app:

- Web server → App server (allowed)
- Database → Web server (blocked)
- User computer → Database (blocked)

Continuous monitoring: Trust expires. If a user's phone leaves the office Wi-Fi, their laptop suddenly accessing the database from a new IP triggers a re-check.

Result: Security adapts to changing conditions.

Key Takeaways

- Automation – Python + netmiko = faster, consistent, auditable network management
- SDN – Separates brain (controller) from muscles (switches); OpenFlow is the language
- Cloud models – IaaS (rent servers), PaaS (deploy code), SaaS (use software)
- VPC – Your own private section of the cloud
- Hybrid cloud – On-prem + cloud for bursting, DR, data sovereignty
- IoT protocols – MQTT (pub/sub), CoAP (UDP), Zigbee (mesh radio)
- IoT architecture – Edge (device), Fog (local), Cloud (central analytics)
- Hierarchical design – Core (backbone), Distribution (policy), Access (endpoints)
- Redundancy – HSRP, EtherChannel, dual PSU; load balancing L4/L7
- Zero Trust – Never trust/always verify, micro-segmentation, continuous monitoring

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Thank You!